US Patent & Trademark Office
Patent Public Search | Text View

United States Patent Application Publication	20250280475
Kind Code	A1
Publication Date	September 04, 2025
Inventor(s)	DENG; Yang et al.

AEROSOL GENERATING DEVICE AND MICROWAVE HEATING ASSEMBLY THEREOF

Abstract

A microwave heating assembly for an aerosol generating device includes: a microwave heating unit includes: an outer conductor unit in a cylindrical shape, an inner conductor unit arranged in the outer conductor unit, and a feed hole communicating an interior of the outer conductor unit with an outside; a radio frequency board; and a microwave feed unit includes: an inner conductor arranged in the feed hole and comprising a feed end and an access end. The feed end is in ohmic contact with an inner side of the outer conductor unit or the inner conductor unit. An access end is in ohmic contact with the radio frequency board.

Inventors: **DENG; Yang (Shenzhen, CN), DU; Jing (Shenzhen, CN), LIANG; Feng (Shenzhen, CN), LAN; Yonghai (Shenzhen, CN)**

Applicant: **SMOORE INTERNATIONAL HOLDINGS LIMITED** (George Town, KY)

Family ID: **91195124**

Appl. No.: **19/213269**

Filed: **May 20, 2025**

Foreign Application Priority Data

CN	202223082483.3	Nov. 21, 2022

Related U.S. Application Data

parent WO continuation PCT/CN2023/105281 20230630 PENDING child US 19213269

Publication Classification

Int. Cl.: **H05B6/80** (20060101); **A24F40/46** (20200101); **H05B6/70** (20060101)

U.S. Cl.:

CPC **H05B6/802** (20130101); **A24F40/46** (20200101); **H05B6/70** (20130101);

Background/Summary

CROSS-REFERENCE TO PRIOR APPLICATION [0001] This application is a continuation of International Patent Application No. PCT/CN2023/105281, filed on Jun. 30, 2023, which claims priority to Chinese Patent Application No. 202223082483.3, filed on Nov. 21, 2022. The entire disclosure of both applications is hereby incorporated by reference herein.

FIELD
[0002] The present invention relates to the field of aerosol generating technologies, and in particular, to an aerosol generating device and a microwave heating assembly thereof.
BACKGROUND
[0003] In the related art, an aerosol generating article generates an aerosol through a microwave heating-type aerosol generating device. However, the microwave heating-type aerosol generating device generally includes a microwave feed unit, and a microwave heating unit and a radio frequency board that are connected by using the microwave feed unit.
[0004] The microwave feed unit is generally a radio frequency connector assembly, and the radio frequency connector assembly includes a male radio frequency terminal and a female radio frequency terminal. The female radio frequency terminal is embedded on the microwave heating unit, the male radio frequency terminal is soldered onto the radio frequency board, and the male radio frequency terminal is connected to the female radio frequency terminal, thereby achieving a good connection between the microwave heating unit and the radio frequency board.
[0005] However, the radio frequency connector assembly in the related art has at least the following disadvantages:
[0006] 1. There are two types of radio frequency terminals: a male radio frequency terminal and a female radio frequency terminal exist. Connection is inconvenient, assembly time is relatively long, reliability is relatively low, and an insertion loss is also increased.
[0007] 2. An radio frequency terminal has a high price. Because there are two types of radio frequency terminals, costs are relatively increased.
SUMMARY
[0008] In an embodiment, the present invention provides a microwave heating assembly for an aerosol generating device, the microwave heating assembly comprising: a microwave heating unit, comprising: an outer conductor unit in a cylindrical shape, an inner conductor unit arranged in the outer conductor unit, and a feed hole communicating an interior of the outer conductor unit with an outside; a radio frequency board; and a microwave feed unit, comprising: an inner conductor arranged in the feed hole and comprising a feed end and an access end, wherein the feed end is in ohmic contact with an inner side of the outer conductor unit or the inner conductor unit, and wherein an access end is in ohmic contact with the radio frequency board.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0009] Subject matter of the present disclosure will be described in even greater detail below based on the exemplary figures. All features described and/or illustrated herein can be used alone or combined in different combinations. The features and advantages of various embodiments will become apparent by reading the following detailed description with reference to the attached

drawings, which illustrate the following:
[0010] FIG. **1** is a schematic structural diagram of a radio frequency board by disassembling a microwave heating assembly according to Embodiment 1 of the present invention;
[0011] FIG. **2** is a longitudinal structural cross-sectional view of the microwave heating assembly according to Embodiment 1 of the present invention;
[0012] FIG. **3** is a longitudinal structural cross-sectional view of a microwave heating unit in a disassembled state according to Embodiment 1 of the present invention;
[0013] FIG. **4** is a schematic structural diagram of a microwave feed unit according to Embodiment 1 of the present invention;
[0014] FIG. **5** is a longitudinal structural cross-sectional view of the microwave feed unit in FIG. **4**;
[0015] FIG. **6** is a longitudinal structural cross-sectional view of the microwave feed unit in FIG. **4** in a disassembled state;
[0016] FIG. **7** is a schematic structural diagram of a microwave heating unit and a microwave feed unit in an assembled state according to Embodiment 2 of the present invention;
[0017] FIG. **8** is a longitudinal structural cross-sectional view of the microwave heating unit and the microwave feed unit in FIG. **7** in the assembled state;
[0018] FIG. **9** is a longitudinal structural cross-sectional view of a microwave heating unit in a disassembled state according to Embodiment 2 of the present invention;
[0019] FIG. **10** is a schematic structural diagram of an inner conductor unit according to Embodiment 2 of the present invention;
[0020] FIG. **11** is a schematic structural diagram of a radio frequency board by disassembling a microwave heating assembly according to Embodiment 3 of the present invention;
[0021] FIG. **12** is a schematic structural diagram of a microwave heating unit and a microwave feed unit in an assembled state according to Embodiment 3 of the present invention;
[0022] FIG. **13** is a longitudinal structural cross-sectional view of the microwave heating unit and the microwave feed unit in FIG. **12** in the assembled state; and
[0023] FIG. **14** is a longitudinal structural cross-sectional view of the microwave heating unit and the microwave feed unit in FIG. **12** in a disassembled state.

DETAILED DESCRIPTION

[0024] In an embodiment, the present invention provides an improved aerosol generating device and a microwave heating assembly thereof.
[0025] In an embodiment, the present invention provides for constructing a microwave heating assembly, used in an aerosol generating device, including: [0026] a microwave heating unit, including: [0027] an outer conductor unit in a cylindrical shape, an inner conductor unit arranged in the outer conductor unit, and a feed hole communicating the interior of the outer conductor unit with the outside; [0028] a radio frequency board; and [0029] a microwave feed unit, including: [0030] an inner conductor arranged in the feed hole and including a feed end and an access end, where the feed end is in ohmic contact with the inner side of the outer conductor unit or the inner conductor unit, and the access end is in ohmic contact with the radio frequency board.
[0031] In some embodiments, the microwave feed unit further includes a connecting conductor arranged at the periphery of an opening of the feed hole and protruding toward the outside; and the microwave heating unit is connected to the radio frequency board by using the connecting conductor.
[0032] In some embodiments, the microwave feed unit further includes an outer conductor; and the outer conductor includes: [0033] a mounting portion, including a first surface, a second surface opposite to the first surface, and a first through hole running through the first surface and the second surface; the first surface is relatively away from the outer conductor unit; and the connecting conductor is fitted to the first surface; and [0034] an embedded portion, configured in a cylindrical shape and embedded in the feed hole, where the end of the embedded portion close to the mounting portion is fitted to the second surface, and a hollow channel of the embedded portion

is in communication with the first through hole.
[0035] In some embodiments, the connecting conductor is integrally fitted to the first surface; or the connecting conductor is in ohmic contact with the first surface.
[0036] In some embodiments, at least one pair of connecting conductors are provided, and are symmetrically distributed on two opposite sides of an opening of the first through hole.
[0037] In some embodiments, the outer conductor further includes a snap portion arranged at the outer periphery of the embedded portion, the snap portion is engaged on the inner wall surface of the feed hole, and the snap portion is in ohmic contact with the outer conductor unit.
[0038] In some embodiments, the snap portion includes a snap ring sleeved at the outer periphery of the embedded portion.
[0039] In some embodiments, the feed hole includes: [0040] a first hole segment close to the inner conductor unit, where the inner diameter of the first hole segment fits the outer diameter of the embedded portion; and [0041] a second hole segment coaxially connected to the first hole segment, where the inner diameter of the second hole segment is greater than the inner diameter of the first hole segment; [0042] and the shape of the inner wall of the second hole segment fits the snap portion.
[0043] In some embodiments, the microwave feed unit further includes a dielectric layer between the outer conductor and the inner conductor, where the dielectric layer is arranged in the through hole and/or the hollow channel.
[0044] In some embodiments, the inner conductor is of a straight-line, needle-like structure, and is coaxially arranged in the first through hole and the hollow channel.
[0045] In some embodiments, the access end extends out of the first through hole, and is flush with the surface of the connecting conductor away from the microwave heating unit.
[0046] In some embodiments, a first spacing is provided between a feedpoint of the radio frequency board and the first surface, and the size of the first spacing is not less than 0.5 mm; and
[0047] a second spacing is provided between the feedpoint of the radio frequency board and the wall surface of the connecting conductor facing the inner conductor, and the size of the second spacing is not less than 0.5 mm.
[0048] In some embodiments, the connecting conductor is integrally fitted to the outer surface of the microwave heating unit, or the connecting conductor is in ohmic contact with the outer surface of the microwave heating unit.
[0049] In some embodiments, at least one pair of connecting conductors are provided, and are symmetrically distributed on two opposite sides of the opening of the feed hole.
[0050] In some embodiments, the inner wall surface of the feed hole forms a circular columnar channel.
[0051] In some embodiments, the microwave feed unit further includes a dielectric layer arranged in the feed hole, the outer diameter of the dielectric layer fits the hole size of the feed hole, and the dielectric layer is between the inner wall surface of the feed hole and the outer wall surface of the inner conductor.
[0052] In some embodiments, the inner conductor is of a straight-line, needle-like structure, and is coaxial with the feed hole.
[0053] In some embodiments, the access end extends out of the feed hole, and is flush with the surface of the connecting conductor away from the microwave heating unit.
[0054] In some embodiments, a third spacing is provided between a feedpoint of the radio frequency board and the outer surface of the microwave heating unit, and the size of the third spacing is not less than 0.5 mm; and
[0055] a fourth spacing is provided between the feedpoint of the radio frequency board and the wall surface of the connecting conductor facing the inner conductor, and the size of the fourth spacing is not less than 0.5 mm.
[0056] In some embodiments, the connecting conductor is further provided with an avoidance

portion recessed along the wall surface of the connecting conductor facing the inner conductor, and the avoidance portion is formed at a position that is of the wall surface of the connecting conductor facing the inner conductor and that is opposite to the inner conductor.
[0057] In some embodiments, the outer conductor unit is cylindrical and includes a first end, a second end, and a cavity between the first end and the second end; and
[0058] the inner conductor unit is coaxially arranged in the cavity and includes a first fixed end and a first free end, the first fixed end is connected to the second end, and the first free end extends toward the first end.
[0059] In some embodiments, the outer conductor unit includes a conductor side wall in a cylindrical shape, and the conductor side wall includes a first open end and a second open end that are opposite.
[0060] In some embodiments, the inner conductor unit includes:
[0061] a conductor column, including a second fixed end and a second free end; and the second fixed end is coaxially connected to the inner side of the outer conductor unit, and the second free end extends toward the first open end.
[0062] In some embodiments, the outer conductor unit further includes a conductor end wall, and the conductor end wall integrally blocks at the second open end; and
[0063] the conductor column includes a first column portion and a second column portion that are coaxially connected, and the second column portion is embedded in the conductor end wall and is in ohmic contact with the conductor end wall; and the first column portion extends from the end of the second column portion close to the first open end toward the first open end.
[0064] In some embodiments, the conductor side wall includes a first cylinder segment and a second cylinder segment that are coaxially connected, the first cylinder segment is relatively away from the second end of the outer conductor unit, and the inner diameter of the first cylinder segment is less than the inner diameter of the second cylinder segment; and
[0065] the conductor column includes a third column portion and a fourth column portion that are coaxially connected, the fourth column portion is embedded in the second cylinder segment, and the outer surface of the fourth column portion is attached to the inner wall surface of the second cylinder segment and is in ohmic contact with the second cylinder segment; and the diameter of the third column portion is less than the diameter of the first cylinder segment, and the third column portion is located in the first cylinder segment.
[0066] In some embodiments, the microwave heating unit further includes a pin configured to fix the fourth column portion in the second cylinder segment; and the pin passes through the outer peripheral side wall of the second cylinder segment, and is inserted into the fourth column portion.
[0067] In some embodiments, a second bump is arranged on the second cylinder segment for limiting the embedding of the fourth column portion in the circumferential direction, and the second bump protrudes inward along the inner wall surface of the second cylinder segment; and the fourth column portion is provided with a through groove matching the second bump and recessed inward along the outer peripheral side wall of the fourth column portion.
[0068] In some embodiments, a first bump is arranged at a position on the conductor side wall close to the second open end; and
[0069] the feed hole radially runs through the conductor side wall and the first bump, and is formed in the conductor side wall and the first bump.
[0070] In some embodiments, the feed hole runs through the conductor end wall in the direction parallel to the axial direction of the outer conductor unit and is formed on the conductor end wall.
[0071] In some embodiments, the feed hole runs through the fourth column portion in the direction parallel to the axial direction of the outer conductor unit and is formed in the fourth column portion.
[0072] In some embodiments, a first insertion hole extending in the radial direction of the conductor column is provided on the outer peripheral wall of the conductor column, and the first

insertion hole is opposite to the feed hole; and

[0073] the feed end is inserted into the first insertion hole, and is in ohmic contact with the conductor column.

[0074] In some embodiments, the inner conductor unit further includes an extending portion fitted to the outer surface of the conductor column, and the extending portion extends outward in the direction perpendicular to the axis direction of the conductor column and is in ohmic contact with the feed end.

[0075] In some embodiments, the extending portion is provided with a second insertion hole for inserting the feed end, and an opening of the second insertion hole is opposite to the feed hole.

[0076] In some embodiments, the inner conductor unit further includes:

[0077] a conductor disk, coaxially connected to the second free end, where the diameter of the conductor disk is greater than the diameter of the conductor column and less than the inner diameter of the cavity.

[0078] In some embodiments, the inner conductor unit further includes:

[0079] a probe device in an elongated shape, where one end of the probe device is embedded in the conductor disk, and the other end of the probe device extends toward the first open end.

[0080] In some embodiments, the microwave heating assembly further includes an accommodating base mounted on the first open end, the accommodating base includes an accommodating portion configured to accommodate an aerosol generating article, and the accommodating portion is located in the cavity.

[0081] In the present invention, an aerosol generating device is further constructed, including a battery assembly and the microwave heating assembly. The battery assembly is electrically connected to the radio frequency board.

[0082] Implementation of the present invention has the following beneficial effects: In the present invention, the two ends of the inner conductor are respectively directly connected to the radio frequency board and the microwave heating unit, thereby reducing a component quantity of the radio frequency connector assembly, reducing costs, and improving reliability of a connection between the microwave heating unit and the radio frequency board.

[0083] List of reference numerals: first microwave heating assembly **100**; [0084] first microwave heating unit **1**; first microwave feed unit **2**; radio frequency board **3**; [0085] first outer conductor unit **11**; first inner conductor unit **12**; first accommodating base **13**; first feed hole **14**; [0086] first end **111**; second end **112**; first conductor side wall **113**; first conductor end wall **114**; first cavity **115**; first bump **116**; sixth side surface **1161**; mounting hole **117**; first conductor column **121**; first conductor disk **122**; first probe device **123**; first column portion **1211**; second column portion **1212**; first insertion hole **1213**; first accommodating portion **131**; first fixing portion **132**; positioning rib **133**; support rib **134**; air inlet gap **135**; accommodating cavity **1311**; second through hole **1321**; first hole segment **141**; second hole segment **142**; first snap slot **143**; [0087] first outer conductor **21**; first inner conductor **22**; first dielectric layer **23**; first connecting conductor **24**; mounting portion **211**; embedded portion **212**; snap portion **213**; first surface **2111**; second surface **2112**; first through hole **2113**; first needle segment **221**; second needle segment **222**; access end **2211**; feed end **2221**; first side surface **241**; second side surface **242**; [0088] second microwave heating assembly **100***a;* [0089] second microwave heating unit **1***a;* [0090] second outer conductor unit **11***a*; second inner conductor unit **12***a*; second accommodating base **13***a*; second feed hole **14***a*; second conductor side wall **113***a*; perforation **118***a*; second bump **119***a*; second cavity **115***a*; first cylinder segment **1131***a*; second cylinder segment **1132***a*; step surface **1133***a*; second conductor column **121***a*; second conductor disk **122***a*; second probe device **123***a*; third column portion **1211***a*; fourth column portion **1212***a*; second insertion hole **1213***a*; extending portion **1214***a*; through groove **1215***a;* [0091] third microwave heating assembly **100***b;* [0092] second microwave feed unit **2***b*; third feed hole **14***b*; second connecting conductor **24***b*; second inner conductor **22***b*; second dielectric layer **23***b*; third side surface **241***b*; fourth side surface **242***b*; fifth side surface **243***b*;

avoidance portion **2431***b*; third needle segment **221***a*; and fourth needle segment **222***a*.
[0093] To provide a clearer understanding of the technical features, objectives, and effects of the present invention, specific implementations of the present invention are described in detail with reference to the accompanying drawings. In the following descriptions of this application, it should be understood that orientation or position relationships indicated by the terms such as "front", "rear", "on", "below", "left", "right", "longitudinal", "latitudinal", "vertical", "horizontal", "top", "bottom", "inside", "outside", "head", and "tail" are based on orientation or position relationships shown in the accompanying drawings, and the orientation or position relationships are used only for ease of describing of the technical solution, rather than indicating that the mentioned apparatus or element needs to have a particular orientation or constructed and operated in a particular orientation. Therefore, such terms should not be construed as a limitation to the present invention.
[0094] It should be further noted that unless otherwise explicitly specified and defined, terms such as "mounted", "connected", "connection", "fixed", and "arranged" should be understood in a broad sense. For example, a connection may be a fixed connection, a detachable connection, or an integral connection; the connection may be a mechanical connection or an electrical connection; or the connection may be a direct connection, an indirect connection through an intermediate medium, internal communication between two elements, or an interaction relationship between two elements. When an element is being "above" or "below" another element, the element can be "directly" or "indirectly" located above the another element, or one or more intermediate elements may exist. The terms such as "first", "second", and "third" are merely intended for case of describing of the technical solution, and shall not be understood as indicating or implying relative significance or implicitly indicating the number of indicated technical features. Therefore, a feature defined by the terms such as "first", "second", and "third" may explicitly or implicitly include one or more of the features. A person of ordinary skill in the art may understand the specific meanings of the foregoing terms in the present invention according to specific situations.
[0095] In the following descriptions, for the purpose of description rather than limitation, specific details such as specific system structures, and technologies are proposed to thoroughly understand the embodiments of the present invention. However, it should be clear to a person skilled in the art that the present invention may also be implemented in other embodiments without these specific details. In other cases, detailed descriptions of well-known systems, apparatuses, circuits, and methods are omitted, so that the present invention is described without being obscured by unnecessary details.
[0096] According to the present invention, an aerosol generating device is constructed. The aerosol generating device can heat an aerosol generating article by using microwaves for atomization to generate an aerosol, for a user to inhale. In some embodiments, the aerosol generating article is a solid-state aerosol generating article such as a processed plant leaf product. It may be understood that, in some other embodiments, the aerosol generating article may be a liquid aerosol generating article.
[0097] Referring to FIG. **1**, in Embodiment 1, the aerosol generating device may include a first microwave heating assembly **100** and a battery assembly. The first microwave heating assembly **100** includes a first microwave heating unit **1**, a first microwave feed unit **2**, and a radio frequency board **3**. In this embodiment, the battery assembly provides electric energy for the radio frequency board **3**, the radio frequency board **3** can generate microwaves, and feed the microwaves to the first microwave heating unit **1** by using the first microwave feed unit **2**, so that the first microwave heating unit **1** forms a microwave field inside, and the microwave field can act on an aerosol generating article, thereby implementing microwave heating on the aerosol generating article.
[0098] As shown in FIG. **1**, the first microwave heating unit **1** is approximately a circular column in overall shape. Certainly, the first microwave heating unit **1** is not limited to being in a shape of the circular column, or may be in another shape such as a rectangular column or an elliptical column. The first microwave heating unit **1** may include a first outer conductor unit **11**, a first inner

conductor unit **12**, and a first accommodating base **13**, and further includes a first feed hole **14** (referring to FIG. **2**) that communicates the interior of the first outer conductor unit **11** with the outside (where the outside refers to an external environment of the first microwave heating unit **1**).
[0099] As shown in FIG. **2**, the first outer conductor unit **11** is in a cylindrical shape, and has a first end **111** and a second end **112** that are opposite. The first inner conductor unit **12** is arranged coaxially in the first outer conductor unit **11**, and is configured to adjust a resonance frequency and microwave distribution inside the first outer conductor unit **11**. One end (namely, a first fixed end) of the first inner conductor unit **12** is connected to the second end **112** of the first outer conductor unit **11**, forming a short-circuit end of the first microwave heating unit **1**; and the other end (namely, the first free end) of the first inner conductor unit **12** extends toward the first end **111** of the first outer conductor unit **11**, and is not in contact with the inner wall surface of the first outer conductor unit **11**, forming an open-circuit end of the first microwave heating unit **1**. The first accommodating base **13** is detachably mounted at the first end **111** of the first outer conductor unit **11**, the partial structure of the first accommodating base **13** is arranged in the first outer conductor unit **11**, and can define an accommodating cavity **1311** in the first outer conductor unit **11** for accommodating a lower structure of the aerosol generating article. The accommodating cavity **1311** is located in an area in which the microwave field is mainly formed.
[0100] In this embodiment, the first outer conductor unit **11** may be integrally formed by using a conductive metal material. The metal material is preferably aluminum alloy or copper. Certainly, the first outer conductor unit **11** is not limited to being integrally formed by using a conductive material, and may be formed by coating the inner wall surface of a non-conductive cylinder with a first conductive coating. The first conductive coating may be made of gold, silver, conductive metal oxide, or the like. Preferably, the first conductive coating is a silver coating or a gold coating.
[0101] As shown in FIG. **2** and FIG. **3**, the first outer conductor unit **11** may include a first conductor side wall **113** and a first conductor end wall **114** that are conductive. The first conductor side wall **113** may be in a cylindrical shape and includes a first open end and a second open end that are arranged oppositely. The first conductor end wall **114** is integrally closed at the second open end, to form a closed end of the first outer conductor unit **11** (namely, the second end **112** of the first outer conductor unit **11**). The first open end forms an open end of the first outer conductor unit **11** (namely, the first end **111** of the first outer conductor unit **11**). The first conductor end wall **114** and the first conductor side wall **113** together define a first cavity **115**. The first cavity **115** is a semi-closed circular columnar channel. The aerosol generating article may extend into the first cavity **115** from the first open end.
[0102] The first conductor end wall **114** is further provided with a mounting hole **117** axially running through the first conductor end wall **114**. The mounting hole **117** is configured for the bottom end of the first inner conductor unit **12** to mount therein.
[0103] The first outer conductor unit **11** further includes a first bump **116**. The first bump **116** is rectangular, radially protrudes outward from a position on the first conductor side wall **113** close to the second open end, and is integrally formed on the outer peripheral wall surface of the first conductor side wall **113**. The first bump **116** includes a sixth side surface **1161** away from the first conductor side wall **113**, and the sixth side surface **1161** directly faces the radio frequency board **3**.
[0104] As shown in FIG. **3**, the first feed hole **14** is configured for embedding the first microwave feed unit **2** therein. The first feed hole **14** radially runs through the first bump **116** and the first conductor side wall **113**, and openings of the first feed hole **14** are respectively formed on the inner peripheral wall of the first conductor side wall **113** and the sixth side surface **1161** of the first bump **116**.
[0105] As shown in FIG. **3**, the first inner conductor unit **12** may include a first conductor column **121**, a first conductor disk **122** arranged above the first conductor column **121**, and a first probe device **123** with an end embedded in the first conductor disk **122** and the first conductor column **121**. Preferably, the axes of the first conductor column **121**, the first conductor disk **122**, the first

probe device **123**, and the first outer conductor unit **11** coincide with each other. It may be understood that, fed microwaves are transmitted to the first probe device **123** by using the first conductor column **121** and the first conductor disk **122**, and a microwave field is formed at the periphery of the first probe device **123**.

[0106] In this embodiment, the first conductor column **121** may be integrally formed by using a conductive metal material, and preferably, aluminum alloy or copper. Certainly, the first conductor column **121** is not limited to being integrally formed by using a conductive material, and may alternatively be formed by coating the outer surface of a non-conductor with a second conductive coating. The second conductive coating is preferably a silver coating or a gold coating.

[0107] The first conductor column **121** is configured in a step-like circular columnar shape. It may be understood that the first conductor column **121** is not limited to the shape of the circular column, and may alternatively be in another shape such as a rectangular column, an elliptical column, or an irregular column. The first conductor column **121** may include a first column portion **1211** and a second column portion **1212** that are integrally connected. The bottom end (namely, a second fixed end) of the second column portion **1212** is coaxially embedded in the mounting hole **117** of the first conductor end wall **114**, and is in ohmic contact with the first conductor end wall **114**. The diameter of the first column portion **1211** is greater than the diameter of the second column portion **1212**, and the first column portion **1211** extends from the top end of the second column portion **1212** toward the first open end. The top end (namely, a second free end) of the first column portion **1211** is located below the first accommodating base **13**.

[0108] A first insertion hole **1213** radially extending along the interior of the first column portion **1211** is provided on the outer peripheral wall surface of the first column portion **1211**. The first insertion hole **1213** is a blind hole, an opening of the first insertion hole **1213** faces the first feed hole **14**, and is configured for inserting the end of the first inner conductor **22** of the first microwave feed unit **2** into the first insertion hole **1213**, so that reliable and good ohmic contact is formed between the first inner conductor **22** and the first conductor column **121**. Certainly, the first insertion hole **1213** is not a necessary part in this embodiment, and is applied to this embodiment as an optional solution. The first insertion hole **1213** is configured for making a connection relationship between the first inner conductor **22** and the first conductor column **121** more reliable. When the first insertion hole **1213** is not provided, the end of the first inner conductor **22** of the first microwave feed unit **2** may directly abut against the outer peripheral wall surface of the first column portion **1211**, to form ohmic contact.

[0109] In this embodiment, the first conductor disk **122** may be integrally formed by using a conductive metal material, and preferably, aluminum alloy or copper. Certainly, the first conductor disk **122** is not limited to being integrally formed by using a conductive material, and may alternatively be formed by coating the outer surface of a non-conductor with a third conductive coating. The third conductive coating is preferably a silver coating or a gold coating.

[0110] The first conductor disk **122** is in the shape of a disk, and the diameter of the first conductor disk **122** is greater than the diameter of the first column portion **1211** of the first conductor column **121** and less than the inner diameter of the first outer conductor unit **11**. The first conductor disk **122** is integrally fitted to the top end of the first column portion **1211**. Certainly, the first conductor disk **122** may alternatively be soldered onto the top end of the first column portion **1211**, and is in ohmic contact with the first column portion **1211**. It may be understood that, the first conductor disk **122** is not a necessary part in this embodiment, and is applied to this embodiment as an optional solution. The first conductor disk **122** may increase the inductance and the capacitance of the first conductor disk **122**, and reduce the resonance frequency, so that the size of the first cavity **115** can be further reduced. When the first conductor disk **122** is not provided, microwave heating can still be implemented by using the first conductor column **121** and the first probe device **123** with the end embedded in the first conductor column **121**.

[0111] In this embodiment, the first probe device **123** may include a first probe configured to adjust microwave field distribution and a microwave feed frequency. The first probe is in an elongated shape, and the lower portion of the first probe may be fixedly or detachably embedded in the first conductor disk **122** and the first conductor column **121**, to form good ohmic contact with the first conductor disk **122** and the first conductor column **121**. The upper end of the first probe extends upward and extends into the accommodating cavity **1311**.

[0112] Preferably, the shape of the upper end portion of the first probe may include one of a plane, a sphere, an ellipsoid, a cone, or a truncated cone. The truncated cone is preferred because the truncated cone can enhance a local field strength, thereby increasing an atomization speed of the aerosol generating article.

[0113] Optionally, the first probe may be integrally formed by using a conductive metal material, and preferably, stainless steel, aluminum alloy, or copper. Certainly, the first probe is not limited to being integrally formed by using a conductive material, and may alternatively be formed by coating the outer surface of a non-conductor with a fourth conductive coating. The fourth conductive coating is preferably a silver coating or a gold coating.

[0114] The first probe device **123** may further include a temperature measurement element arranged in the first probe, and the temperature measurement element is configured to monitor the internal temperature of the aerosol generating article arranged in the first accommodating base **13**. It may be understood that when temperature measurement is not required, the probe may be a solid structure; and when temperature measurement is required, the probe may be a hollow probe.

[0115] As shown in FIG. **3**, the first accommodating base **13** may include a first accommodating portion **131** and a first fixing portion **132** integrally connected to the first accommodating portion **131**. The first accommodating portion **131** is configured to define the accommodating cavity **1311**. The first fixing portion **132** is configured to axially block the first open end of the first conductor side wall **113**, and allow the first accommodating portion **131** to extend into the first cavity **115**.

[0116] The first accommodating portion **131** may be cylindrical, and the outer diameter of the first accommodating portion **131** may be less than the inner diameter of the first conductor side wall **113**. The first accommodating portion **131** defines an axial accommodating cavity **1311**. The first fixing portion **132** may be annular and is coaxially connected to the first accommodating portion **131**. The first fixing portion **132** may coaxially block the first open end of the first conductor side wall **113**. The first fixing portion **132** includes an axial second through hole **1321** connecting the accommodating cavity **1311** to the outside, and the aerosol generating article may be inserted into the accommodating cavity **1311** through the second through hole **1321**.

[0117] Preferably, the first accommodating base **13** further includes several elongated positioning ribs **133**. The positioning ribs **133** are uniformly spaced circumferentially on the wall surfaces of the accommodating cavity **1311** and/or the second through hole **1321**. Each positioning rib **133** extends in the direction parallel to the axis of the first accommodating base **13**. The positioning ribs **133** may be configured to clamp the aerosol generating article inserted into the accommodating cavity **1311** and/or the through hole. In addition, a first air inlet channel longitudinal extending is formed between every two adjacent positioning ribs **133**, so that air in the environment is conveniently drawn to the bottom of the aerosol generating article, and then enters the aerosol generating article, to carry the aerosol of the aerosol generating article generated through microwave heating.

[0118] Preferably, an air inlet gap **135** may further be provided below the accommodating cavity **1311**, to prevent blocked air flow due to the bottom end surface of the aerosol generating article being completely covered. In this embodiment, the first accommodating portion **131** includes the inner side end surface abutting against the bottom end surface of the aircraft-forming article, and support ribs **134** that are uniformly spaced and radially distributed are arranged on the inner side end surface. The inner side end surface supports the aerosol generating article by using the support ribs **134**. In addition, the support ribs **134** form several radial second air inlet channels (namely, the air inlet gaps **135**). The second air inlet channels respectively communicate with the first air inlet

channels, so that environment air is drawn into the bottom of the aerosol generating article, and then enters the aerosol generating article, to carry the aerosol generated by microwaves heating.
[0119] Optionally, the first accommodating base **13** may be made of a high-temperature-resistant material with a low dielectric loss, to reduce a proportion of microwave energy absorbed by the first accommodating base **13** and improve a heating (carbonization) effect of the aerosol generating article.
[0120] In this embodiment, the radio frequency board **3** may be arranged in the circumferential direction of the first microwave heating unit **1**. The radio frequency board **3** may include a microstrip, a feedpoint connected to the microstrip, and a pair of pads located around the feedpoint. The pair of pads are symmetrically distributed on two opposite sides of the feedpoint. It should be noted that the radio frequency board **3** belongs to the existing technology. For a specific construction, refer to the existing technology. Details are not described herein again.
[0121] As shown in FIG. **4**, the first microwave feed unit **2** may be a coaxial connector, which is inserted from the first feed hole **14**, and is partially embedded in the first feed hole **14**. The first microwave feed unit **2** includes a first outer conductor **21**, a first inner conductor **22**, a first dielectric layer **23**, and a pair of first connecting conductors **24**.
[0122] In this embodiment, as shown in FIG. **4** to FIG. **6**, the first outer conductor **21** includes a mounting portion **211**, an embedded portion **212**, and a snap portion **213**. The mounting portion **211** is of a rectangular sheet structure, and is located between an opening of the first feed hole **14** away from the first inner conductor unit **12** and the first connecting conductor **24**. The mounting portion **211** includes a first surface **2111**, a second surface **2112** opposite to the first surface **2111**, and a first through hole **2113** running through the first surface **2111** and the second surface **2112**. The first surface **2111** is relatively away from the first outer conductor unit **11**. The embedded portion **212** is cylindrical and is embedded in the first feed hole **14**. The end of the embedded portion **212** close to the mounting portion **211** is fitted to the second surface **2112** of the mounting portion **211**, and a hollow channel of the embedded portion **212** is in communication with the first through hole **2113**, The snap portion **213** may include an annular snap ring, and the snap ring is sleeved at the outer periphery of the embedded portion **212**. When the embedded portion **212** is arranged in the first feed hole **14**, the snap ring may be engaged on the inner wall surface of the first feed hole **14**, so that good and reliable ohmic contact can be formed between the first outer conductor **21** and the first outer conductor unit **11**.
[0123] It may be understood that the first feed hole **14** may include a first hole segment **141** and a second hole segment **142** that are coaxially connected. The first hole segment **141** is close to the first inner conductor unit **12**, and the hole size of the first hole segment **141** fits the outer diameter of the embedded portion **212**. The outer wall surface of the embedded portion **212** may be attached to the inner wall surface of the first hole segment **141**. The inner diameter of the second hole segment **142** is greater than the inner diameter of the first hole segment **141**, and the shape of the inner wall of the second hole segment **142** fits the shape of the snap ring. In this embodiment, the inner wall surface of the second hole segment **142** forms a circumferential first snap slot **143**, and the first microwave feed unit **2** is engaged in the first snap slot **143** by using the snap portion **213** of the first microwave feed unit **2**, to be reliably fixed in the first feed hole **14**.
[0124] Preferably, to prevent the first outer conductor **21** and the connecting conductor from interfering with a radio frequency signal at a position of the feedpoint, a first spacing needs to be maintained between the feedpoint of the radio frequency board **3** and the first surface **2111** of the mounting portion **211**. The size of the first spacing is not less than 0.5 mm.
[0125] As shown in FIG. **4**, the pair of first connecting conductors **24** are symmetrically distributed on two opposite sides at the periphery of an opening of the first through hole **2113**, a symmetry line of the pair of first connecting conductors **24** is parallel to the axis of the first outer conductor unit **11**, and projection edges of the pair of first connecting conductors **24** close to the first through hole **2113** on the first surface **2111** are tangent to an edge of the opening of the first through hole **2113**.

Preferably, the first connecting conductor **24** is of a square sheet structure and includes a first side surface **241** and a second side surface **242** that are opposite. The first side surface **241** is relatively away from the first outer conductor unit **11** and is configured to be soldered onto the pad of the radio frequency board **3**. The second side surface **242** is integrally fitted to the first surface **2111** of the mounting portion **211**. It may be understood that the radio frequency board **3** and the first microwave heating unit **1** can be conducted by soldering the pair of first connecting conductors **24** onto the pad of the radio frequency board **3**. The shape of the first connecting conductors **24** fits the shape of the pad, and a quantity of the first connecting conductors **24** corresponds to a quantity of the pads.

[0126] Preferably, to prevent the first outer conductor **21** and the connecting conductor from interfering with a radio frequency signal at the position of the feedpoint, a second spacing needs to be maintained between the feedpoint of the radio frequency board **3** and the wall surface of the first connecting conductor **24** facing the first inner conductor **22**. The size of the second spacing is not less than 0.5 mm.

[0127] As shown in FIG. **2**, FIG. **4**, and FIG. **6**, the first inner conductor **22** is partially arranged in the first outer conductor **21**. The first inner conductor **22** is of a straight-line, needle-like structure, and may include a first needle segment **221** and a second needle segment **222** coaxially and integrally connected to the first needle segment **221**. The first needle segment **221** is relatively away from the first outer conductor unit **11**, and the end of the first needle segment **221** away from the second needle segment **222** is the access end **2211**, configured to extend out of the first through hole **2113** to come into contact with the feedpoint of the radio frequency board **3**. The diameter of the access end **2211** fits the diameter of the feedpoint, and the access end **2211** is located outside the first through hole **2113** and is flush with the first side surface **241** of the first connecting conductor **24**. The diameter of the second needle segment **222** is greater than the diameter of the first needle segment **221**, and the end of the second needle segment **222** away from the first needle segment **221** is a feed end **2221**, configured to extend into the first cavity **115** and the first insertion hole **1213** of the first inner conductor unit **12**, and abut against the inner wall surface of the first insertion hole **1213**, to come into ohmic contact with the first inner conductor unit **12**.

[0128] It may be understood that the first inner conductor **22** is not limited to being in a straight-line shape. In some other embodiments, the first inner conductor **22** may alternatively be in an L shape, and includes a first segment perpendicular to the axis of the first outer conductor unit **11** and a second segment parallel to the axis of the first outer conductor unit **11**. The first segment is integrally connected to the second segment. The end (namely, the access end **2211**) of the first segment away from the second segment is connected to the feedpoint of the radio frequency board **3**. The end (namely, the feed end **2221**) of the second segment away from the first segment is in ohmic contact with the inner wall surface of the first outer conductor unit **11** (for example, the inner wall surface of the first conductor end wall **114**).

[0129] The first dielectric layer **23** may include a pair of insulators of an arc structure. The pair of insulators are between the first outer conductor **21** and the first inner conductor **22** and are symmetrically arranged in the circumferential direction of the first inner conductor **22**, and inner concave side surfaces of the pair of insulators both face the outer peripheral surface of the first inner conductor **22**.

[0130] It may be understood that, in Embodiment 1, by using the feature of the first outer conductor unit **11** being made of a conductive metal material or being formed by coating the inner wall surface of the non-conductive cylinder with the first conductive coating, the first feed hole **14** is provided on the first outer conductor unit **11** (it should be noted that a specific opening size conforms to a radio frequency connector design specification/an impedance matching specification), and a partial structure of the radio frequency connector assembly is transferred to the first outer conductor unit **11**, so that the first outer conductor unit **11** has a function of an outer conductor of a radio frequency connector. In addition, an improved male radio frequency terminal (namely, the first microwave feed unit **2**) may be directly embedded in the first feed hole **14**, one end of the male radio frequency terminal is directly soldered onto the pad of the radio frequency board **3**, and the other end of the male radio frequency terminal is directly in ohmic contact with the first microwave heating unit **1**, and does not need to be combined with and connected to the first microwave heating unit **1** by using a female radio frequency terminal, thereby omitting the female radio frequency terminal. Such construction can simplify a structure and reduce a volume of the radio frequency connector assembly, which is beneficial to a miniaturized design of the aerosol generating device, and reduces costs. In addition, an assembly speed can be improved, connection reliability can be improved, and insertion loss can be reduced.

[0131] Referring to FIG. **7** together, the second microwave heating assembly **100***a* in Embodiment 2 of the present invention is shown. The second microwave heating assembly **100***a* is improved based on Embodiment 1. A difference between the second microwave heating assembly **100***a* and the first microwave heating assembly **100** in Embodiment 1 lies in that: The first microwave heating unit **1** in Embodiment 1 is replaced with a second microwave heating unit **1***a*.

[0132] As shown in FIG. **7** and FIG. **8**, the second microwave heating unit **1***a* is approximately a circular column in overall shape, may include a second outer conductor unit **11***a*, a second inner conductor unit **12***a*, and a second accommodating base **13***a*, and further includes a second feed hole **14***a* communicating the interior of the second outer conductor unit **11***a* with the outside.

[0133] In this embodiment, the second outer conductor unit **11***a* is in a step-like cylindrical shape, and may be integrally formed by using a conductive metal material or by coating the inner wall surface of a non-conductive cylinder with the first conductive coating. As shown in FIG. **9**, the second outer conductor unit **11***a* may include a conductive second conductor side wall **113***a*. The second conductor side wall **113***a* may be cylindrical, including a first cylinder segment **1131***a* and a second cylinder segment **1132***a* that are in communication longitudinally. The first cylinder segment **1131***a* is located above the second cylinder segment **1132***a*, the upper end of the first cylinder segment **1131***a* is of an open structure, and the lower end of the first cylinder segment **1131***a* is integrally connected to the upper end of the second cylinder segment **1132***a*. The lower end of the second cylinder segment **1132***a* is also of an open structure, and the inner diameter of the second cylinder segment **1132***a* is greater than the inner diameter of the first cylinder segment **1131***a*, so that a step surface **1133***a* is formed between the first cylinder segment **1131***a* and the second cylinder segment **1132***a*.

[0134] The second outer conductor unit **11***a* further includes a perforation **118***a* and a second bump **119***a* that are arranged on the second cylinder segment **1132***a*. The perforation **118***a* radially runs through the outer peripheral side wall of the second cylinder segment **1132***a*, and is formed on the second cylinder segment **1132***a*. The second bump is square, and is formed by radially protruding inward along the inner wall surface of the second cylinder segment **1132***a*. The perforation **118***a* and the second bump **119***a* are distributed at an interval in the circumferential direction of the second cylinder segment **1132***a*, and are both configured to cooperate with the second inner conductor unit **12***a*, to quickly fix the second inner conductor unit **12***a* in the second outer conductor unit **11***a*.

[0135] As shown in FIG. **9** and FIG. **10**, the second inner conductor unit **12***a* may include a second conductor column **121***a*, a second conductor disk **122***a* arranged above the second conductor column **121***a*, and a second probe device **123***a* with an end embedded in the second conductor disk **122***a* and the second conductor column **121***a*. Preferably, the axes of the second conductor column **121***a*, the second conductor disk **122***a*, the second probe device **123***a*, and the second outer conductor unit **11***a* coincide with each other.

[0136] In this embodiment, the second conductor column **121***a* may be integrally formed by using a conductive metal material or by coating the outer surface of a non-conductor with the second conductive coating. The second conductor column **121***a* is in a shape of a step-like circular column, and may include a third column portion **1211***a* and a fourth column portion **1212***a* that are integrally connected. The fourth column portion **1212***a* is in a shape of a circular column, and the diameter of the fourth column portion **1212***a* fits the inner diameter of the second cylinder segment **1132***a*. The third column portion **1211***a* is in a shape of a square column, and the outer diameter of the third column portion **1211***a* is less than the inner diameter of the first cylinder segment **1131***a* and less than the diameter of the fourth column portion **1212***a*. The third column portion **1211***a* is located above the fourth column portion **1212***a*, and is integrally fitted to the top surface of the fourth column portion **1212***a*. The second conductor column **121***a* may be embedded into the inner wall of the second conductor side wall **113***a* from below the second conductor side wall **113***a*. The fourth column portion **1212***a* is accommodated in the second cylinder segment **1132***a*, the outer peripheral wall surface of the fourth column portion **1212***a* is attached to the inner peripheral wall surface of the second cylinder segment **1132***a*, and the top surface of the fourth column portion **1212***a* abuts against the step surface **1133***a* between the first cylinder segment **1131***a* and the second cylinder segment **1132***a*, so that good ohmic contact is formed between the fourth column portion **1212***a* and the second conductor side wall **113***a*. It may be understood that, the fourth column portion **1212***a* is configured to block the lower end of the second conductor side wall **113***a*, and a semi-closed second cavity **115***a* is formed between the fourth column portion **1212***a* and the first cylinder segment **1131***a*. The third column portion **1211***a* is located in the first cylinder segment **1131***a*, and the top end of the third column portion **1211***a* is configured to be connected to the second conductor disk **122***a*.

[0137] The second conductor column **121***a* further includes a through groove **1215***a* and a hole provided on the outer peripheral side wall of the fourth column portion **1212***a*. In an assembly process in which the second conductor column **121***a* is mounted on the second conductor side wall **113***a*, the through groove **1215***a* needs to be inserted into the second bump **119***a*, thereby successfully embedding the fourth column portion **1212***a* into the second cylinder segment **1132***a*. The through groove **1215***a* and the second bump **119***a* are designed to position the perforation **118***a* and the hole, so that the perforation **118***a* and the hole can quickly come into communication. In this case, a pin may be inserted through the through hole **118***a* into the hole, and is fixed in the through hole **118***a* and the hole, thereby fixing the second conductor column **121***a* in the second conductor side wall **113***a*. Optionally, the through groove **1215***a* is radially recessed inward along the outer peripheral side wall of the fourth column portion **1212***a* and formed, and the shape of the through groove **1215***a* fits the shape of the second bump **119***a*. The hole is also provided on the outer peripheral side wall of the fourth column portion **1212***a*, and is formed by extending inward along the outer peripheral side wall of the fourth column portion **1212***a*. The hole and the through groove **1215***a* are distributed at an interval on the outer peripheral side wall of the fourth column portion **1212***a*, and a position of the hole relative to the through groove **1215***a* corresponds to a position of the perforation **118***a* relative to the second bump **119***a*.

[0138] The second conductor column **121***a* further includes an extending portion **1214***a* radially extending outward along the outer peripheral side wall of the third column portion **1211***a*. The extending portion **1214***a* is configured to be in ohmic contact with the feed end **2221** of the first inner conductor **22** of the first microwave feed unit **2**. Preferably, a second insertion hole **1213***a* is further provided on the wall surface of the extending portion **1214***a* facing the second feed hole **14***a*. The second insertion hole **1213***a* is a blind hole, and an opening of the second insertion hole **1213***a* is opposite to the second feed hole **14***a* and is configured for inserting the feed end **2221** of the first inner conductor **22** into the second insertion hole **1213***a*, so that the feed end **2221** abuts against the inner wall surface of the second insertion hole **1213***a*, and good and reliable ohmic contact is formed between the first inner conductor **22** and the second inner conductor unit **12***a*.

[0139] As shown in FIG. **8** to FIG. **10**, the second feed hole **14***a* is configured for embedding the first microwave feed unit **2** therein. The second feed hole **14***a* is located at the periphery of the third column portion **1211***a*, runs through the upper end surface and the lower end surface of the fourth column portion **1212***a* in the direction parallel to the axial direction of the second outer conductor unit **11***a*, and is formed in the fourth column portion **1212***a*. It may be understood that for a specific configuration of the second feed hole **14***a*, reference may be made to the first feed hole **14**, and details are not described herein again.

[0140] In this embodiment, the radio frequency board **3** may be arranged below the second microwave heating unit **1***a*.

[0141] It should be noted that for specific configurations and connection relationships of the second conductor disk **122***a*, the second probe device **123***a*, and the second accommodating base **13***a* in the second inner conductor unit **12***a*, reference may be made to the first conductor disk **122**, the first probe device **123**, and the first accommodating base **13** in Embodiment 1, and details are not described herein again.

[0142] It may be understood that, in Embodiment 2, similarly, by using the feature of the second inner conductor unit **12***a* being made of a conductive metal material or being formed by coating the inner wall surface of a non-conductive cylinder with the second conductive coating, the second feed hole **14***a* is provided on the second inner conductor unit **12***a*, and a partial structure of the radio frequency connector assembly is transferred to the second inner conductor unit **12***a*, so that the second inner conductor unit **12***a* has the function of the outer conductor of the radio frequency connector. For other specific functions, reference may be made to Embodiment 1, and details are not described herein again. It can be seen from FIG. **7** that, the second microwave heating unit **1***a* in Embodiment 2 is relatively longer and thinner than the first microwave heating unit **1** in Embodiment 1, further facilitating a miniaturized design of the aerosol generating device.

[0143] Referring to FIG. **11** together, the third microwave heating assembly **100***b* in Embodiment 3 of the present invention is shown. The third microwave heating assembly **100***b* is improved based on Embodiment 1. A difference between the third microwave heating assembly **100***b* and the first microwave heating assembly **100** in Embodiment 1 lies in that: The first microwave feed unit **2** in Embodiment 1 is replaced with the second microwave feed unit **2***b*. In addition, correspondingly, to match the second microwave feed unit **2***b*, the first feed hole **14** in Embodiment 1 is replaced with a third feed hole **14***b*.

[0144] In this embodiment, the third feed hole **14***b* is a circular columnar channel with a uniform inner diameter, and is formed in the first bump **116** of the first outer conductor unit **11** and the first conductor side wall **113**.

[0145] As shown in FIG. **14**, the second microwave feed unit **2***b* includes a pair of second connecting conductors **24***b*, a second inner conductor **22***b*, and a second dielectric layer **23***b*.

[0146] As shown in FIG. **11** to FIG. **13**, the pair of second connecting conductors **24***b* are symmetrically distributed on two opposite sides at the periphery of an opening of the third feed hole **14***b*, and a symmetry line of the pair of second connecting conductors **24***b* is parallel to the axis of the first outer conductor unit **11**. The second connecting conductor **24***b* is of a square sheet structure, and may include a third side surface **241***b* (equivalent to the first side surface **241** of the first connecting conductor **24**) and a fourth side surface **242***b* (equivalent to the second side surface **242** of the first connecting conductor **24**) that are opposite. The third side surface **241***b* is relatively away from the second outer conductor unit **11***a*, and is configured to be soldered onto the pad of the radio frequency board **3**. The fourth side surface **242***b* is fitted to the sixth side surface **1161** of the first bump **116** (which may be integrally fitted, or may be soldered onto the sixth side surface **1161** of the first bump **116**). The first outer conductor unit **11** and the radio frequency board **3** may be soldered by using the pair of second connecting conductor **24***b*, so that good ohmic contact is formed between the first outer conductor unit **11** and the radio frequency board **3**.

[0147] The second connecting conductor **24***b* further includes a fifth side surface **243***b* connected between the third side surface **241***b* and the fourth side surface **242***b*, and the fifth side surface **243***b* faces the second inner conductor **22***b*. Preferably, an avoidance portion **2431***b* is provided at a position on the fifth side surface **243***b* opposite to the second inner conductor **22***b*. The avoidance portion **2431***b* is an arc groove, is recessed on the fifth side surface **243***b* in the direction away from the second inner conductor **22***b* and formed, and extends from the third side surface **241***b* toward the fourth side surface **242***b*. An objective of the avoidance portion **2431***b* is to maintain a spacing between the feedpoint of the radio frequency board **3** and the fifth side surface **243***b*, to prevent the second connecting conductor **24***b* from interfering with a radio frequency signal at the position of the feedpoint. Preferably, the spacing is not less than 0.5 mm. In addition, a third spacing may further be maintained between the feedpoint of the radio frequency board **3** and the sixth side surface **1161** of the first bump **116**, to prevent the first bump **116** from interfering with a radio frequency signal at the position of the feedpoint. The size of the third spacing is not less than 0.5 mm.

[0148] As shown in FIG. **14**, the second inner conductor **22***b* is partially arranged in the third feed hole **14***b*. The second inner conductor **22***b* is of a straight-line, needle-like structure, and may include a third needle segment **221***b* and a fourth needle segment **222***b* integrally connected to the third needle segment **221***b*. The third needle segment **221***b* is equivalent to the first needle segment **221** of the structure of the first inner conductor **22**, the fourth needle segment **222***b* is equivalent to the second needle segment **222** of the structure of the first inner conductor **22**. For a specific structure and connection relationship of the second inner conductor **22***b*, reference may be made to the first inner conductor **22**, and details are not described herein again.

[0149] It may be understood that, the second inner conductor **22***b* is not limited to being in a straight-line shape, and may alternatively be in an L shape. For details, refer to related descriptions in Embodiment 1.

[0150] The second dielectric layer **23***b* may include a cylindrical insulating cylinder. The insulating cylinder is sleeved at the outer periphery of the second inner conductor **22***b*, the outer diameter of the insulating cylinder fits the inner diameter of the third feed hole **14***b*, and the outer peripheral wall surface of the insulating cylinder is attached to the inner wall surface of the third feed hole **14***b*, to fix the second inner conductor **22***b* to be coaxially located in the third feed hole **14***b*.

[0151] It may be understood that in Embodiment 3, based on Embodiment 1, the structure of the radio frequency connector assembly is further simplified, the male radio frequency terminal is canceled, the second connecting conductor **24***b* is directly fitted to the first bump **116** of the first outer conductor unit **11**, and the second inner conductor **22***b* is supported by using the second dielectric layer **23***b* to be coaxially arranged in the third feed hole **14***b*, thereby implementing a microwave feed function. Such construction further simplifies the structure, further reduces the volume of the radio frequency connector assembly, and further facilitates a miniaturized design of the aerosol generating device, and further reduces costs.

[0152] While the invention has been illustrated and described in detail in the drawings and foregoing description, such illustration and description are to be considered illustrative or exemplary and not restrictive. It will be understood that changes and modifications may be made by those of ordinary skill within the scope of the following claims. In particular, the present invention covers further embodiments with any combination of features from different embodiments described above and below. Additionally, statements made herein characterizing the invention refer to an embodiment of the invention and not necessarily all embodiments.

[0153] The terms used in the claims should be construed to have the broadest reasonable interpretation consistent with the foregoing description. For example, the use of the article “a” or “the” in introducing an element should not be interpreted as being exclusive of a plurality of elements. Likewise, the recitation of “or” should be interpreted as being inclusive, such that the recitation of “A or B” is not exclusive of “A and B,” unless it is clear from the context or the foregoing description that only one of A and B is intended. Further, the recitation of “at least one of A, B and C” should be interpreted as one or more of a group of elements consisting of A, B and C, and should not be interpreted as requiring at least one of each of the listed elements A, B and C, regardless of whether A, B and C are related as categories or otherwise. Moreover, the recitation of “A, B and/or C” or “at least one of A, B or C” should be interpreted as including any singular entity from the listed elements, e.g., A, any subset from the listed elements, e.g., A and B, or the entire list of elements A, B and C.

Claims

1. A microwave heating assembly for an aerosol generating device, the microwave heating assembly comprising: a microwave heating unit, comprising: an outer conductor unit in a cylindrical shape, an inner conductor unit arranged in the outer conductor unit, and a feed hole communicating an interior of the outer conductor unit with an outside; a radio frequency board; and a microwave feed unit, comprising: an inner conductor arranged in the feed hole and comprising a feed end and an access end, wherein the feed end is in ohmic contact with an inner side of the outer conductor unit or the inner conductor unit, and wherein an access end is in ohmic contact with the radio frequency board.
2. The microwave heating assembly of claim 1, wherein the microwave feed unit comprises a connecting conductor arranged at a periphery of an opening of the feed hole and protruding toward the outside, and wherein the microwave heating unit is connected to the radio frequency board using the connecting conductor.
3. The microwave heating assembly of claim 2, wherein the microwave feed unit comprises an outer conductor; and the outer conductor comprises: a mounting portion comprising a first surface, a second surface opposite to the first surface, and a first through hole running through the first surface and the second surface, the first surface being away from the outer conductor unit, the connecting conductor being fitted to the first surface; and an embedded portion in a cylindrical shape and embedded in the feed hole, an end of the embedded portion close to the mounting portion being fitted to the second surface, a hollow channel of the embedded portion being in communication with the first through hole.
4. The microwave heating assembly of claim 3, wherein the connecting conductor is integrally fitted to the first surface, or wherein the connecting conductor is in ohmic contact with the first surface.
5. The microwave heating assembly of claim 3, wherein at least one pair of connecting conductors is provided, the at least one pair of connecting conductors being symmetrically distributed on two opposite sides of an opening of the first through hole.
6. The microwave heating assembly of claim 3, wherein the outer conductor comprises a snap portion arranged at an outer periphery of the embedded portion, wherein the snap portion is engaged on an inner wall surface of the feed hole, and wherein the snap portion is in ohmic contact with the outer conductor unit.
7. The microwave heating assembly of claim 6, wherein the snap portion comprises a snap ring sleeved at an outer periphery of the embedded portion.
8. The microwave heating assembly of claim 6, wherein the feed hole comprises: a first hole segment close to the inner conductor unit, an inner diameter of the first hole segment fitting an outer diameter of the embedded portion; and a second hole segment coaxially connected to the first hole segment, an inner diameter of the second hole segment being greater than an inner diameter of the first hole segment, wherein a the shape of the inner wall of the second hole segment fits the snap portion.
9. The microwave heating assembly of claim 3, wherein the microwave feed unit comprises a dielectric layer between the outer conductor and the inner conductor, and wherein the dielectric layer is arranged in the through hole and/or the hollow channel.
10. The microwave heating assembly of claim 3, wherein the inner conductor comprises a straight-line, needle-like structure, and is coaxially arranged in the first through hole and the hollow channel.
11. The microwave heating assembly of claim 3, wherein the access end extends out of the first

through hole, and is flush with a surface of the connecting conductor away from the microwave heating unit.
12. The microwave heating assembly of claim 11, wherein a first spacing is provided between a feedpoint of the radio frequency board and the first surface, and a size of the first spacing is not less than 0.5 mm, and wherein a second spacing is provided between the feedpoint of the radio frequency board and a wall surface of the connecting conductor facing the inner conductor, and a size of the second spacing is not less than 0.5 mm.
13. The microwave heating assembly of claim 2, wherein the connecting conductor is integrally fitted to an outer surface of the microwave heating unit, or wherein the connecting conductor is in ohmic contact with an outer surface of the microwave heating unit.
14. The microwave heating assembly of claim 13, wherein at least one pair of connecting conductors are provided, the at least one pair of connecting conductors being symmetrically distributed on two opposite sides of the opening of the feed hole.
15. The microwave heating assembly of claim 2, wherein an inner wall surface of the feed hole forms a circular columnar channel.
16. The microwave heating assembly of claim 15, wherein the microwave feed unit comprises a dielectric layer arranged in the feed hole, wherein an outer diameter of the dielectric layer fits a hole size of the feed hole, and wherein the dielectric layer is between an inner wall surface of the feed hole and an outer wall surface of the inner conductor.
17. The microwave heating assembly of claim 2, wherein the inner conductor comprises a straight-line, needle-like structure, and is coaxial with the feed hole.
18. The microwave heating assembly of claim 17, wherein the access end extends out of the feed hole, and is flush with a surface of the connecting conductor away from the microwave heating unit.
19. The microwave heating assembly of claim 13, wherein a third spacing is provided between a feedpoint of the radio frequency board and an outer surface of the microwave heating unit, and a size of the third spacing is not less than 0.5 mm, and wherein a fourth spacing is provided between the feedpoint of the radio frequency board and a wall surface of the connecting conductor facing the inner conductor, and a size of the fourth spacing is not less than 0.5 mm.
20. The microwave heating assembly of claim 12, wherein the connecting conductor is provided with an avoidance portion recessed along a wall surface of the connecting conductor facing the inner conductor, and wherein the avoidance portion is formed at a position of the wall surface of the connecting conductor facing the inner conductor and that is opposite to the inner conductor.
21. The microwave heating assembly of claim 2, wherein the outer conductor unit is cylindrical and comprises a first end, a second end, and a cavity between the first end and the second end, and wherein the inner conductor unit is coaxially arranged in the cavity and comprises a first fixed end and a first free end, the first fixed end is connected to the second end, and the first free end extends toward the first end.
22. The microwave heating assembly of claim 21, wherein the outer conductor unit comprises a conductor side wall in a cylindrical shape, and wherein the conductor side wall comprises a first open end and a second open end that are opposite.
23. The microwave heating assembly of claim 22, wherein the inner conductor unit comprises: a conductor column comprising a second fixed end and a second free end, wherein the second fixed end is coaxially connected to an inner side of the outer conductor unit, and wherein the second free end extends toward the first open end.
24. The microwave heating assembly of claim 23, wherein the outer conductor unit comprises a conductor end wall, and the conductor end wall integrally blocks at the second open end, wherein the conductor column comprises a first column portion and a second column portion that are coaxially connected, wherein the second column portion is embedded in the conductor end wall and is in ohmic contact with the conductor end wall, and wherein the first column portion extends

from an end of the second column portion close to the first open end toward the first open end.
25. The microwave heating assembly of claim 23, wherein the conductor side wall comprises a first cylinder segment and a second cylinder segment that are coaxially connected, wherein the first cylinder segment is away from the second end of the outer conductor unit, wherein an inner diameter of the first cylinder segment is less than an inner diameter of the second cylinder segment, wherein the conductor column comprises a third column portion and a fourth column portion that are coaxially connected, wherein the fourth column portion is embedded in the second cylinder segment, wherein an outer surface of the fourth column portion is attached to an inner wall surface of the second cylinder segment and is in ohmic contact with the second cylinder segment, wherein a diameter of the third column portion is less than a diameter of the first cylinder segment, and the third column portion is located in the first cylinder segment.
26. The microwave heating assembly of claim 25, wherein the microwave heating unit comprises a pin configured to fix the fourth column portion in the second cylinder segment, and wherein the pin passes through an outer peripheral side wall of the second cylinder segment, and is inserted into the fourth column portion.
27. The microwave heating assembly of claim 26, wherein a second bump is arranged on the second cylinder segment so as to limit embedding of the fourth column portion in a circumferential direction, wherein the second bump protrudes inward along an inner wall surface of the second cylinder segment, and wherein the fourth column portion is provided with a through groove matching the second bump and recessed inward along an outer peripheral side wall of the fourth column portion.
28. The microwave heating assembly of claim 24, wherein a first bump is arranged at a position on the conductor side wall close to the second open end, and wherein the feed hole radially runs through a conductor side wall and the first bump, and is formed in the conductor side wall and the first bump.
29. The microwave heating assembly of claim 24, wherein the feed hole runs through the conductor end wall in the direction parallel to an axial direction of the outer conductor unit and is formed on the conductor end wall.
30. The microwave heating assembly of claim 25, wherein the feed hole runs through the fourth column portion in a direction parallel to an axial direction of the outer conductor unit and is formed in the fourth column portion.
31. The microwave heating assembly of claim 28, wherein a first insertion hole extending in a radial direction of the conductor column is provided on an outer peripheral wall of the conductor column, wherein the first insertion hole is opposite to the feed hole, and wherein the feed end is inserted into the first insertion hole, and is in ohmic contact with the conductor column.
32. The microwave heating assembly of claim 29, wherein the inner conductor unit comprises an extending portion fitted to an outer surface of the conductor column, and wherein the extending portion extends outward in a direction perpendicular to an axis direction of the conductor column and is in ohmic contact with the feed end.
33. The microwave heating assembly of claim 32, wherein the extending portion is provided with a second insertion hole for inserting the feed end, and wherein an opening of the second insertion hole is opposite to the feed hole.
34. The microwave heating assembly of claim 23, wherein the inner conductor unit comprises: a conductor disk coaxially connected to the second free end, wherein a diameter of the conductor disk is greater than a diameter of the conductor column and less than an inner diameter of the cavity.
35. The microwave heating assembly of claim 34, wherein the inner conductor unit comprises: a probe device in an elongated shape, wherein one end of the probe device is embedded in the conductor disk, and wherein an other end of the probe device extends toward the first open end.
36. The microwave heating assembly of claim 22, further comprising: an accommodating base

mounted on the first open end, the accommodating base comprising an accommodating portion configured to accommodate an aerosol generating article, wherein the accommodating portion is located in the cavity.

37. An aerosol generating device, comprising: a battery assembly; the microwave heating assembly of claim, and wherein the battery assembly is electrically connected to the radio frequency board.